SCCI_RH / ANALOG REFERENCE

5 V / ARTIFICIAL SIGNALS / ELECTRICAL TEST LOADS ONLY

RECORDING + BUFFERED MIDPOINT

PRECISION SINK + POLARITY SWITCH

J1
+5V 1
GND 2
5V INPUT

J2
SIG_IN 1
GND 2
MID 3
ARTIFICIAL IN

J3
REC_OUT 1
GND 2
MID 3
ANALOG OUT

PWR_FLAG +5V
PWR_FLAG GND

U1
+5V 8 V+
REC_IN 3 IN_A+
REC_FB 2 IN_A−
IN_A+ OUT_A 1 REC_AMP
MID_RAW 5 IN_B+
MID 6 IN_B−
IN_B+ OUT_B 7 MID
GND 4 V−
OPA2333AID

U2
+5V 8 V+
I_REF_RAW 3 IN_A+
I_REF 2 IN_A−
IN_A+ OUT_A 1 I_REF
I_REF 5 IN_B+
I_SENSE 6 IN_B−
IN_B+ OUT_B 7 GATE_DRV
GND 4 V−
OPA2333AID

Q1
GATE 1 G
D 3 SINK
S 2 I_SENSE
2N7002

Q2
INHIBIT 1 G
D 3 +5V
S 2 SINK
2N7002

U3
+5V 16 VDD
INHIBIT 6 INH
PHASE 11 S_A
PHASE 10 S_B
GND 9 S_C
GND 7 VEE
GND 8 VSS
A0 12 +5V
A1 13 SINK
A 14 SW_A
B0 2 SINK
B1 1 +5V
B 15 SW_B
C0 5 GND
C1 3 GND
C 4 GND
74HC4053D

J4
PHASE 1
INHIBIT 2
+5V 3
GND 4
5V LOGIC

J5
DUT_A 1
DUT_B 2
DUMMY LOAD

PASSIVES / net labels define connectivity

R1 SIG_IN — AC_IN — 10k
C1 AC_IN — REC_IN — 1u
R2 REC_IN — MID — 1M
R3 REC_FB — MID — 1k
R4 REC_AMP — REC_FB — 20k
R5 REC_AMP — REC_OUT — 1k
C2 REC_OUT — GND — 47n

R6 +5V — MID_RAW — 10k
R7 MID_RAW — GND — 10k
C3 MID_RAW — GND — 10u
C5 +5V — GND — 100n
C8 +5V — GND — 10u
R8 +5V — I_REF_RAW — 49.9k
R9 I_REF_RAW — GND — 1.02k

C4 I_REF_RAW — GND — 100n
R10 I_SENSE — GND — 1k
R11 GATE_DRV — GATE — 1k
C6 +5V — GND — 100n
R12 INHIBIT — +5V — 100k
R13 PHASE — GND — 100k
C7 +5V — GND — 100n

R14 SW_A — DUT_A — 12k
R15 SW_B — DUT_B — 12k

OPA2333: OUTA=1, INA−=2, INA+=3, V−=4, INB+=5, INB−=6, OUTB=7, V+=8

PHASE 0: A=+5V, B=sink. PHASE 1: A=sink, B=+5V. INHIBIT 1: isolated.

External 5 V sequencer: 200 us enabled phase 0, inhibit for 50 us, 200 us phase 1, inhibit.

No MCU, ADC, current monitor, patient isolation, implant package or validated safety interlock populated.

R8/R9/R10: 5*(1.02k)/(49.9k+1.02k)/1k = 100.157 uA nominal; calibrate if ever fabricated.

Digital reference only; no body connection

Project SCCI RH

Sheet: /
File: scci_rh_reference.kicad_sch

Title: Project SCCI RH | Artificial−signal analog reference

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/1